

PULSAR ELECTRONIC SOLUTIONS

GRIDx SMART METERING PLATFORM

Tariff Management System & Billing Engine Architecture

Technical & Business Reference Document

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CONFIDENTIAL

Prepared for: Utilities, Municipalities, Government Ministries, Strategic Partners

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1. Executive Summary

The GRIDx Smart Metering Platform is a comprehensive, end-to-end energy management ecosystem designed to serve utilities, municipalities, and energy distributors across Southern Africa. At its core lies a sophisticated tariff management system and billing engine that supports multiple tariff structures, dual billing modes (prepaid and postpaid), real-time energy metering, and full regulatory compliance with Namibian and SADC electricity standards.

This document provides a complete technical and business overview of how tariffs are created, managed, assigned, and enforced across the GRIDx ecosystem, spanning the ESP32-based smart meter firmware, the cloud backend platform, the mobile application, and the web-based administration dashboard.

1.1 Key Capabilities

- 27 pre-configured Windhoek 2024 ECB-compliant tariff categories
- Support for Flat Rate, Block/Tiered, and Time-of-Use (TOU) tariff structures
- Dual billing modes: Prepaid (STS token vending) and Postpaid (monthly invoicing)
- Real-time MQTT-based tariff deployment to smart meters
- Net metering with bidirectional energy tracking and export compensation
- Automated arrears recovery and regulatory levy collection
- IEC 62055-41 STS standard compliance for token generation
- Demand-Side Management (DSM) with automated geyser load control
- Full audit trail with itemised transaction line items
- External API gateway for third-party integration

2. Tariff Management Overview

The GRIDx tariff management system is designed to accommodate the full range of tariff structures used by Namibian electricity distributors, including the 27 official tariff categories defined by the Electricity Control Board (ECB) for the City of Windhoek.

2.1 Tariff Types Supported

2.1.1 Flat Rate Tariffs

A single rate (N\$/kWh) is applied uniformly to all energy consumed regardless of volume or time of day. This is the simplest tariff model, commonly used for residential prepaid meters.

- **Example:** Residential Prepaid (RESPP): N\$2.32/kWh flat rate
- **Use Case:** Standard domestic consumers on prepaid meters
- **Additional Charges:** Capacity charge, demand charge, network access charge (where applicable)

2.1.2 Block / Tiered Tariffs

Consumption is divided into ranges (blocks), each charged at a different rate. This supports inclining block tariffs that encourage energy conservation by charging higher rates for higher consumption.

Block	Range (kWh)	Rate (N\$/kWh)	Example Tariff
Block 1	0 - 200 kWh	1.52	Social Prepaid (Pensioner)
Block 2	201+ kWh	2.32	Social Prepaid (Pensioner)

Block tariff calculation: For a customer consuming 250 kWh under the Social Prepaid (Pensioner) tariff:

- Block 1: 200 kWh x N\$1.52 = N\$304.00
- Block 2: 50 kWh x N\$2.32 = N\$116.00
- Total energy cost: N\$420.00 (before levies and VAT)

2.1.3 Time-of-Use (TOU) Tariffs

Different rates are applied based on the time of day and day of the week. The GRIDx platform implements the official City of Windhoek TOU schedule with three rating periods: Peak, Standard, and Off-Peak.

TOU tariffs are detailed in Section 4 of this document.

2.1.4 Seasonal Tariffs

The GRIDx architecture supports seasonal tariff variations through configurable effective dates on tariff groups. Utilities can create separate tariff configurations for summer and winter periods, with automatic switching based on the effective date field.

2.1.5 Net Metering Tariffs

For prosumers with solar installations, GRIDx provides dedicated net metering tariff categories that support both flat-rate and TOU-based export compensation. Detailed in Section 7.

- Net Metering Flat (NETF): N\$1.24/kWh
- Net Metering TOU (NETT): Peak N\$1.77, Standard N\$1.32, Off-Peak N\$0.88/kWh

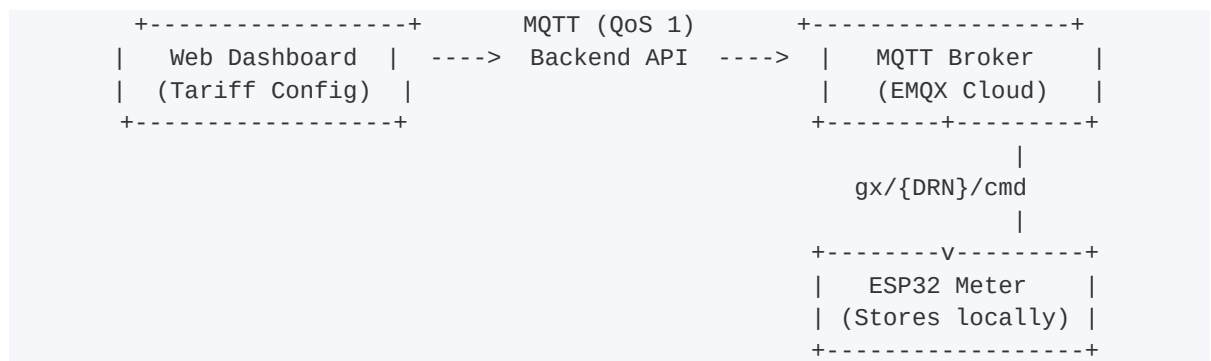
2.2 Tariff Data Architecture

Tariffs are stored across four interconnected database tables:

Table	Purpose	Key Fields
TariffGroups	Master tariff definitions	name, type (Block/Flat/TOU), billingType, rates, charges, effectiveDate
TariffBlocks	Tiered/block ranges and rates	tariffGroupId, rangeLabel, rate, minKwh, maxKwh, period
TariffTOUSchedule	TOU period schedules	tariffGroupId, dayOfWeek, startHour, endHour, period
TariffConfig	System-wide regulatory config	vatRate, fixedCharge, relLevy, ecbLevy, nefLevy, laSurcharge

2.3 Tariff Synchronisation to Meters

Tariff configurations are pushed from the backend to smart meters via the MQTT protocol. The platform uses a publish-subscribe model where each meter subscribes to its command topic (gx/{DRN}/cmd) and receives tariff updates as JSON payloads.



The MQTT command payload for tariff deployment:

- TOU Mode: { type: 'tou_config', mode: 'tou', rates: { peak, standard, off_peak }, schedule: [...], group: 'name' }
- Flat Mode: { type: 'tou_config', mode: 'flat', rate: 2.15, group: 'name' }

The system tracks deployment status per meter in the tariff_update_status table, with states: Pending, Updated, Confirmed, Failed. This enables fleet-wide rollout monitoring.

2.4 Windhoek 2024 ECB Tariff Categories

The platform ships pre-configured with all 27 official Windhoek 2024 ECB tariff categories:

Code	Category	Type	Mode	Rate (N\$/kWh)
GENPP	General Prepaid	Flat	Prepaid	3.48

RESPP	Residential Prepaid	Flat	Prepaid	2.32
SOCSP	Social Services Prepaid	Flat	Prepaid	2.41
SOCPP	Social Prepaid (Pensioner)	Block	Prepaid	1.52 / 2.32
RES20	Residential <=20A	Flat	Postpaid	1.57
EL20	Business <=75A	TOU	Postpaid	Peak: 2.67
INDDK	Industrial Demand	TOU	Postpaid	Demand-based
NETF	Net Metering Flat	Flat	Prepaid	1.24
NETT	Net Metering TOU	TOU	Prepaid	0.88 - 1.77

Note: Table shows selected categories. All 27 ECB categories are pre-loaded in the system.

3. Billing System Overview

The GRIDx billing engine supports both prepaid and postpaid billing models, with full integration into the tariff management system. All financial transactions are atomically processed with complete audit trails.

3.1 Prepaid Billing (Token Vending)

The prepaid system follows the IEC 62055-41 STS standard for token generation and credit management. When a customer purchases electricity, the system calculates all applicable deductions and generates a 20-digit STS token.

3.1.1 Credit Deduction Sequence

Every prepaid token vend follows a strict deduction waterfall:

#	Component	Type	Amount	Basis
1	Value Added Tax (VAT)	Tax	15%	On total amount
2	Fixed Charge	Standing	N\$8.50	Per transaction
3	REL Levy	Regulatory	N\$2.40	Per transaction
4	ECB Levy	Regulatory	N\$0.0212/kWh	Per kWh consumed
5	NEF Levy	Regulatory	N\$0.0160/kWh	Per kWh consumed
6	LA Surcharge	Regulatory	N\$0.1200/kWh	Per kWh consumed
7	Arrears Recovery	Debt	Up to 25%	Of post-VAT amount
8	Energy Credit	Energy	Remainder	Converted to kWh

3.1.2 Worked Example

Customer purchases N\$100.00 on the Residential Prepaid tariff (N\$2.32/kWh):

Calculation Step	Amount (N\$)	Running Total
Total Payment	100.00	100.00
Less VAT (100 - 100/1.15)	-13.04	86.96
Less Fixed Charge	-8.50	78.46
Less REL Levy	-2.40	76.06
Available for Energy	76.06	
Energy: 76.06 / 2.32		32.78 kWh
Less ECB Levy (32.78 x 0.0212)	-0.69	
Less NEF Levy (32.78 x 0.0160)	-0.52	
Less LA Surcharge (32.78 x 0.12)	-3.93	
Net Energy Amount	70.92	30.57 kWh

3.1.3 Token Generation

Upon successful calculation, the system generates a 20-digit numeric STS token, records the transaction atomically (master record + 8-9 line items), and delivers the token via SMS, print receipt, or mobile app notification.

3.2 Postpaid Billing

Postpaid meters accumulate consumption without credit deduction. Bills are generated on a configurable cycle (1st of month, 15th, end of month, or custom day).

3.2.1 Bill Generation

- Consumption (kWh) = MAX(cumulative units) - MIN(cumulative units) during billing period
- Energy charge = consumption x tariff rate
- Fixed charge = N\$8.50 (configurable)
- VAT = (energy charge + fixed charge) x 15%
- Total = energy charge + fixed charge + VAT
- Due date = billing period end + credit days (7/14/30 days configurable)

3.2.2 Payment Recording & Supply Control

Payments are recorded against individual bills with partial payment support. The system tracks:

- Bill status: Generated, Sent, Partial, Paid, Overdue
- Automatic supply disconnection when outstanding amount exceeds turn_off_max_amount
- Automatic supply reconnection when payment brings balance below turn_on_max_amount

3.3 Billing Mode Switching

The GRIDx platform supports seamless switching between prepaid and postpaid billing modes:

3.3.1 Prepaid to Postpaid

- MQTT command sent: { type: 'billing_mode', mode: 'postpaid', action: 'consume_and_switch' }
- Meter consumes remaining prepaid credit
- Credit decrement disabled, relay kept ON
- Mode change logged in MeterModeHistory table

3.3.2 Postpaid to Prepaid

- Final bill auto-generated for usage from month-start to current date
- Outstanding bills flagged (but switch allowed)
- MQTT command: { type: 'billing_mode', mode: 'prepaid', action: 'switch' }
- Meter relay set to OFF until tokens are loaded

3.4 Arrears Recovery

The system supports three arrears recovery modes:

Mode	Behaviour	Configuration
Auto-Deduct	Automatically deducts up to 25% of each token purchase towards	arrearsPercentage: 25%, arrearsThreshold: N\$500

	outstanding arrears	
Manual	Operator manually records arrears payments against customer accounts	Deduction not automatic; tracked in customer record
Disabled	No arrears tracking or deduction	Arrears field remains but is not acted upon

4. Time-of-Use (TOU) Implementation

Time-of-Use tariffs represent the most sophisticated pricing mechanism in the GRIDx platform, enabling utilities to incentivise demand shifting away from peak periods and optimise grid utilisation.

4.1 Rating Periods

Period	Description	Typical Rate
Peak	Highest demand periods; highest tariff rate applied to discourage consumption	N\$2.67/kWh
Standard	Moderate demand periods; intermediate tariff rate	N\$2.17/kWh
Off-Peak	Low demand periods; discounted tariff rate to encourage load shifting	N\$1.68/kWh

4.2 Windhoek 2024 TOU Schedule

4.2.1 Weekday Schedule (Monday - Friday)

Time Range	Period	Rate	Duration	Signal
00:00 - 06:00	Off-Peak	N\$1.68	6 hours	Low demand
06:00 - 07:00	Standard	N\$2.17	1 hour	Ramp-up
07:00 - 10:00	Peak	N\$2.67	3 hours	Morning peak
10:00 - 17:00	Standard	N\$2.17	7 hours	Midday
17:00 - 21:00	Peak	N\$2.67	4 hours	Evening peak
21:00 - 22:00	Standard	N\$2.17	1 hour	Ramp-down
22:00 - 24:00	Off-Peak	N\$1.68	2 hours	Night

4.2.2 Saturday Schedule

Time Range	Period	Rate
00:00 - 07:00	Off-Peak	N\$1.68
07:00 - 12:00	Standard	N\$2.17
12:00 - 18:00	Off-Peak	N\$1.68
18:00 - 20:00	Standard	N\$2.17
20:00 - 24:00	Off-Peak	N\$1.68

4.2.3 Sunday / Public Holiday Schedule

All hours classified as Off-Peak (N\$1.68/kWh) for the full 24-hour period.

4.3 Meter Clock Synchronisation

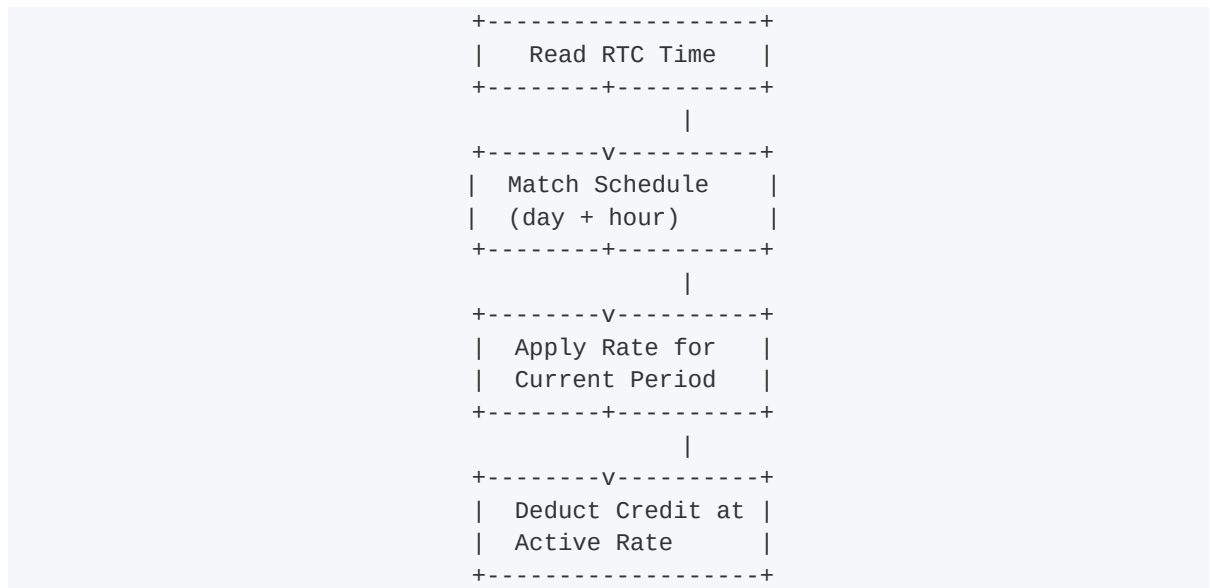
Accurate TOU billing depends on the meter maintaining correct time. The GRIDx system uses multiple clock synchronisation mechanisms:

- NTP Time Sync: ESP32 meters synchronise with NTP servers on every network reconnection
- MQTT Heartbeat: Each telemetry payload includes a record_time (Unix timestamp) validated against server time

- **RTC Backup:** On-board Real-Time Clock (RTC) module maintains time during network outages
- **Drift Correction:** Backend validates meter timestamps; if drift exceeds 30 seconds, a time-sync command is issued

4.4 Automatic Tariff Switching

Once a TOU schedule is pushed to the meter, the meter firmware autonomously manages rate switching:



This approach ensures the meter operates correctly even when the MQTT connection is unavailable, as the entire schedule is stored locally in non-volatile memory.

5. Meter-Side Implementation

The GRIDx smart meter, based on the ESP32 microcontroller platform, implements all tariff and billing logic locally, enabling autonomous operation during network outages.

5.1 Local Tariff Storage

- Tariff configuration stored in SPIFFS (SPI Flash File System) on the ESP32
- Supports up to 10 tariff rate indices (0-9) with labels: Free/Emergency, Lifeline, Standard, Commercial, Industrial, Custom 5-9
- TOU schedule stored as a 168-entry array (7 days x 24 hours)
- Active tariff index selectable via MQTT command or local console

5.2 Credit Deduction Logic (Prepaid)

The meter performs real-time energy accounting:

- Energy consumption measured via precision energy IC at 1-second intervals
- Credit decremented using active tariff rate: $\text{credit_remaining} -= (\text{watts} * \text{time_delta} / 3600) * \text{rate}$
- When credit reaches zero, the supply relay is opened (disconnected)
- Emergency credit mode provides a configurable kWh buffer before full disconnection
- Credit balance transmitted to backend via MQTT telemetry every 30 seconds

5.3 Import/Export Energy Tracking

For net metering installations, the meter independently tracks bidirectional energy flow:

- Import Energy: Accumulated watt-hours drawn from the grid (positive flow)
- Export Energy: Accumulated watt-hours fed back to the grid from solar (negative flow)
- Net Energy: Import minus Export (positive = net consumer, negative = net producer)
- All three values transmitted as a 17-byte binary MQTT packet every 30 seconds

5.4 Offline Operation

The meter is designed for reliable operation during extended network outages:

Feature	Offline Behaviour
Tariff Rates	Continue using last-configured rates from SPIFFS
TOU Schedule	RTC-driven; switches periods autonomously
Credit Deduction	Continues real-time deduction from stored balance
Token Processing	Accepts and processes STS tokens locally
Energy Metering	Cumulative import/export totals stored in flash
Relay Control	Enforces credit limits and load control independently
Data Buffering	Stores readings in ring buffer; uploads on reconnection

5.5 Synchronisation on Reconnection

When MQTT connectivity is restored:

- Buffered telemetry data (power, energy, net energy) is uploaded in order
- Current credit balance and cumulative energy totals are synced
- Any pending tariff updates from the backend are received and applied
- Meter firmware checks for OTA (Over-The-Air) updates
- Time is re-synchronised via NTP

6. Net Metering Implementation

The GRIDx net metering system enables prosumers with solar PV installations to participate in bidirectional energy exchange with the utility grid. The system tracks import and export energy flows in real-time and provides comprehensive analytics.

6.1 Energy Measurement

6.1.1 Import Energy

Energy drawn from the utility grid, measured in watt-hours (Wh). This is standard consumption energy that the customer pays for at the applicable tariff rate.

6.1.2 Export Energy

Energy generated by the customer's solar PV system that exceeds local consumption and is fed back into the grid. Measured in watt-hours (Wh) through the same metering element by detecting reverse power flow.

6.1.3 Net Energy Calculation

Net Energy = Import Energy - Export Energy

- Positive net energy: Customer is a net consumer (paying for import)
- Negative net energy: Customer is a net producer (earning export credit)
- Zero net energy: Perfect balance between production and consumption

6.2 MQTT Data Protocol

Net energy data is transmitted as a 17-byte binary MQTT payload on topic `gx/{DRN}/net_energy`:

Offset	Field	Type	Description
0	Type	uint8	0x07 (net_energy identifier)
1-4	Import Energy	float LE	Cumulative import in Wh
5-8	Export Energy	float LE	Cumulative export in Wh
9-12	Net Energy	float LE	Import minus Export in Wh
13-16	Record Time	uint32 LE	Unix timestamp

6.3 Compensation Tariffs

Export energy is compensated at rates defined in the net metering tariff groups:

Tariff Category	Structure	Compensation Rate
Net Metering Flat (NETF)	Flat Rate	N\$1.24/kWh
Net Metering TOU (NETT)	Time-of-Use	Peak: N\$1.77 / Std: N\$1.32 / Off: N\$0.88

6.4 Customer Scenario Example

Residential customer with 5kW solar PV installation, on Net Metering Flat tariff (N\$1.24/kWh export, N\$2.32/kWh import):

Daily Profile:

- 06:00 - 08:00: Import 4 kWh (no solar, morning usage) = N\$9.28 cost
- 08:00 - 16:00: Export 12 kWh (solar excess fed to grid) = N\$14.88 credit
- 16:00 - 22:00: Import 6 kWh (evening consumption) = N\$13.92 cost
- 22:00 - 06:00: Import 2 kWh (overnight standby) = N\$4.64 cost

Daily Summary:

- Total Import: 12 kWh = N\$27.84
- Total Export: 12 kWh = N\$14.88 credit
- Net Cost: N\$27.84 - N\$14.88 = N\$12.96 (53% saving vs. non-solar)

6.5 Dashboard Reporting

The web dashboard provides real-time net metering analytics:

- Hourly energy flow chart (import vs export bar chart, 24 hours)
- Dark blue summary strip: Total Export, Total Import, Est. Revenue, Active Net Meters
- Period-selectable trends: Daily, Weekly, Monthly, Yearly with ToggleButtonGroup
- Active meters table with live power direction (IMPORTING/EXPORTING/IDLE)
- Per-meter drill-down with hourly and daily breakdowns

6.6 Pre-Computed Analytics

To ensure fast dashboard loading, the backend maintains pre-computed summary tables:

- MeterNetEnergyHourly: UPSERT on each MQTT reading with MIN/MAX import/export per hour
- MeterNetEnergyDaily: Same aggregation at daily level
- Delta calculation: $\text{energy_consumed} = \text{MAX}(\text{cumulative}) - \text{MIN}(\text{cumulative})$ per time bucket

This eliminates expensive full-table scans on raw telemetry data, reducing dashboard load time from seconds to milliseconds.

7. Tariff Assignment to Specific Meters

The GRIDx platform provides flexible tariff assignment mechanisms that enable utilities to apply different pricing structures to individual meters or groups of meters.

7.1 Assignment Methods

7.1.1 Customer-Level Assignment

Each VendingCustomer record contains a tariffGroup field that links the customer to a specific TariffGroup. This is the primary assignment mechanism used during customer registration.

7.1.2 Meter Profile Assignment

The MeterProfileReal table includes a tariff_type field that stores the assigned tariff category. This enables meter-level tariff management independent of the customer record.

7.1.3 Meter Tier System

The MeterBillingConfiguration table includes a meter_tier field (Tier 1, Tier 2, Tier 3) that enables differential billing treatment within the same tariff group.

7.2 Tariff Group Categories

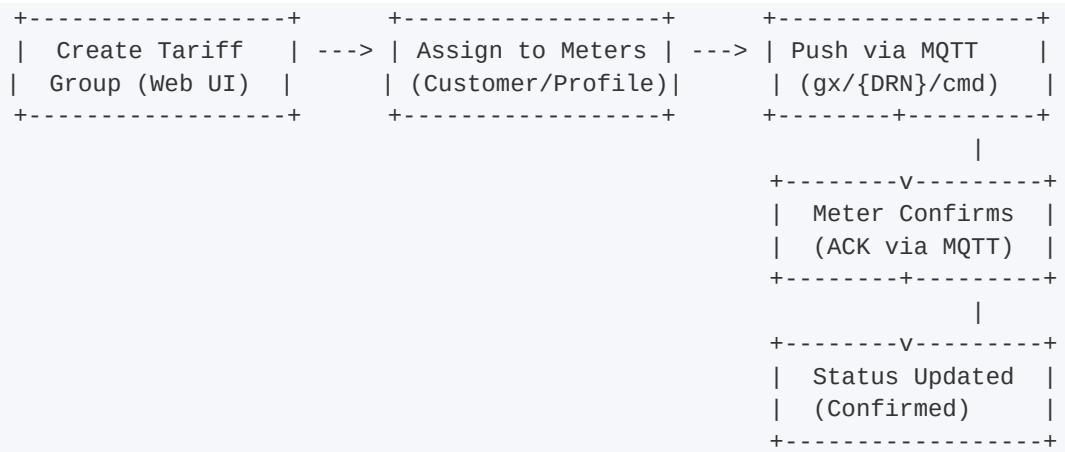
Group Type	Description	Example Tariffs
Residential	Standard domestic consumers on prepaid or postpaid	RESPP, RES20, RES60
Commercial	Business premises including shops and offices	EL20, EL60, COMPP
Industrial	Large power consumers with demand charges	INDDK
Social/Subsidised	Government-subsidised tariffs for pensioners and social services	SOCPP, SOCSP
Net Metering	Prosumers with solar PV installations	NETF, NETT
Municipal	Streetlights, public infrastructure, government buildings	STREET, FLOODLT

7.3 Bulk Tariff Deployment

The platform supports fleet-wide tariff updates:

- Single meter push: POST /vending/tariffs/push/{DRN} with tariffGroup name
- Bulk push: POST /vending/tariffs/push-all pushes to all meters matching a tariff type
- Deployment tracking: Each meter's update status monitored (Pending/Updated/Confirmed/Failed)
- Retry mechanism: Failed deployments can be retried with timestamp tracking

7.4 Tariff Assignment Workflow



8. Application & Web Dashboard Screens

The GRIDx ecosystem provides comprehensive user interfaces across the web dashboard and mobile application for managing tariffs, billing, and net metering operations.

8.1 Web Dashboard

8.1.1 Tariff Management Screen

The Tariffs page (/tariffs) provides three management tabs:

- Prepaid Tariff Settings: 10-slot rate table (indices 0-9), system configuration panel for VAT/levies, and 'Apply to All Prepaid Meters' bulk deployment button
- ECB Tariff Groups: Full CRUD interface for all 27 Windhoek 2024 categories with TOU schedule editor (24x7 visual grid), rate configuration, and per-group meter push
- Postpaid Tariff Settings: Create/edit postpaid tariff configurations with flat or tiered rate structures

8.1.2 Billing & Revenue Screen

The Billing page provides:

- Summary dashboard: Total revenue (prepaid + postpaid), outstanding arrears, meter counts by mode
- Bill generation interface for postpaid meters with configurable billing periods
- Payment recording with partial payment support
- Mode switching interface (prepaid to postpaid and vice versa)
- Revenue charts: Daily/weekly trends, prepaid vs postpaid comparison

8.1.3 Net Metering Dashboard

The Net Metering page (/net-metering) displays:

- Real-time hourly energy flow chart with dark blue summary card
- Period-selectable net energy trends (Daily/Weekly/Monthly/Yearly toggle)
- Active meters table with live status indicators and power direction
- Revenue estimation based on export compensation tariffs

8.1.4 Dashboard Revenue Analytics

The main Dashboard includes a Revenue Analytics chart with a period selector (Hourly/Daily/Weekly/Monthly/Yearly) showing total revenue, peak period revenue, and average revenue with interactive bar charts.

8.1.5 Vending Interface

The Vending page provides:

- Customer lookup with meter number search
- Token generation with real-time tariff-based breakdown calculation
- Amount presets (N\$10, N\$25, N\$50, N\$100, N\$200, N\$500)
- Transaction receipt with itemised deductions

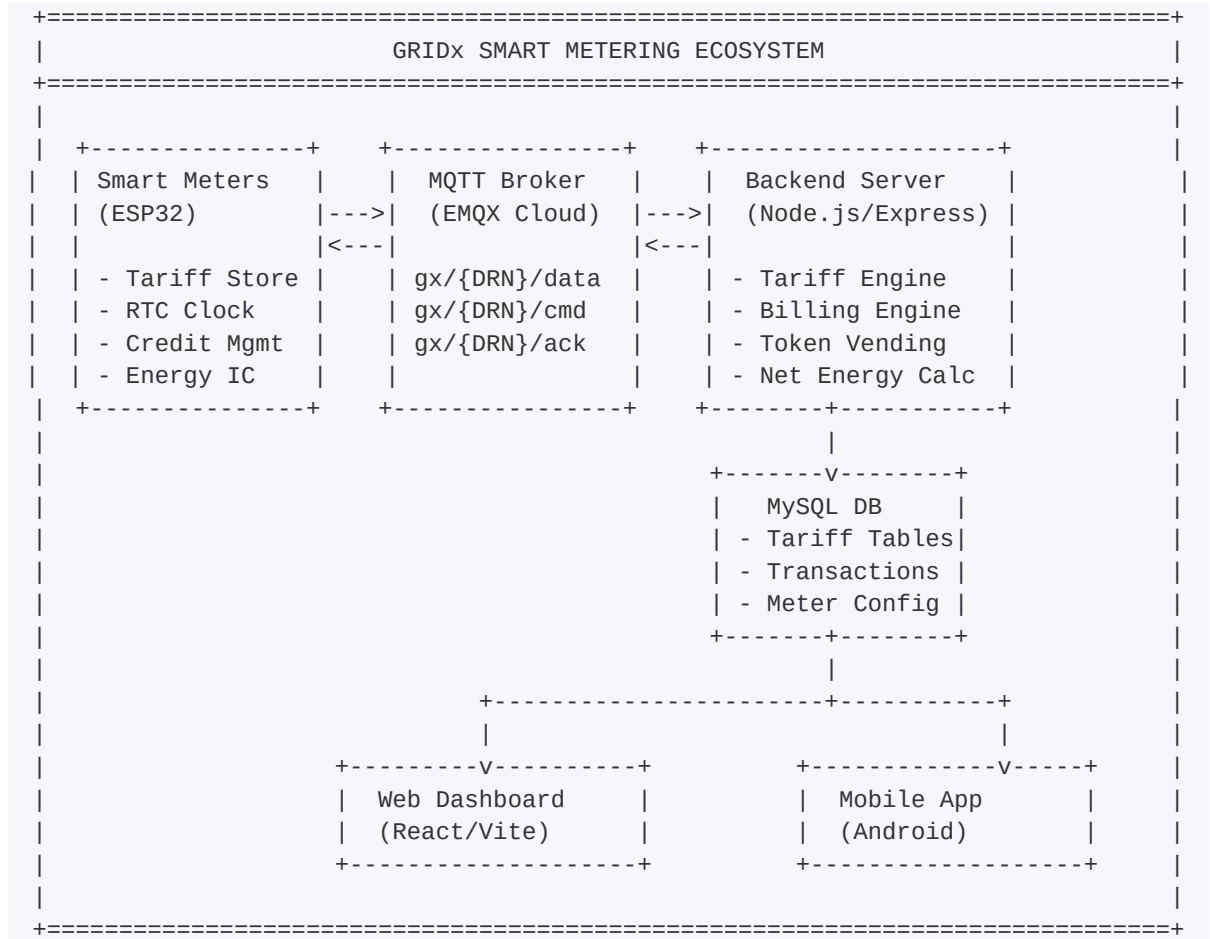
8.2 Mobile Application

The GRIDx Android application provides field-level access to:

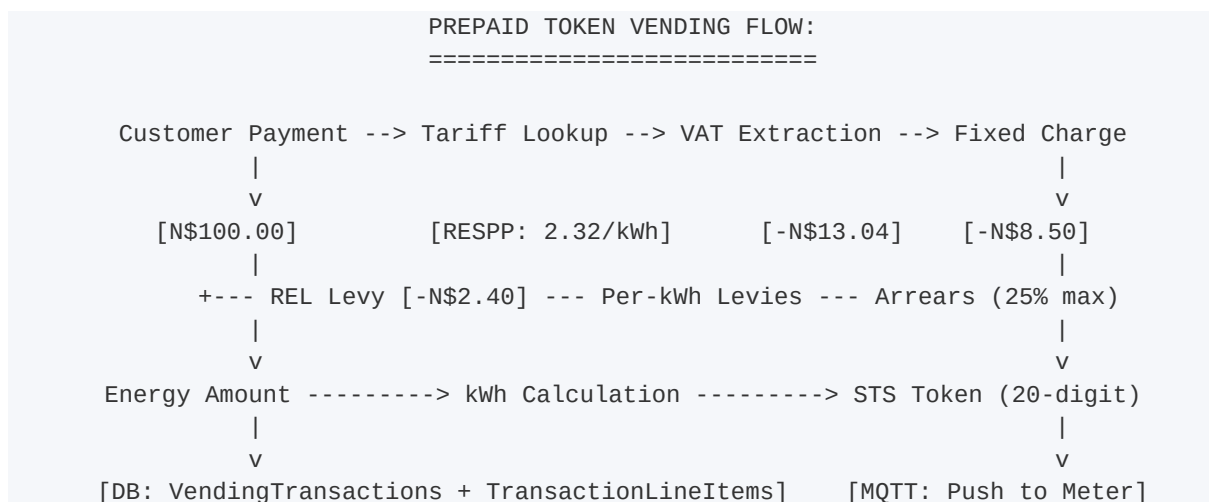
- Real-time power flow visualisation (import/export direction indicators)
- Hourly and daily energy analytics per meter
- STS token entry and processing
- Meter health monitoring and diagnostics
- Net metering energy balance display

9. System Architecture & Data Flow

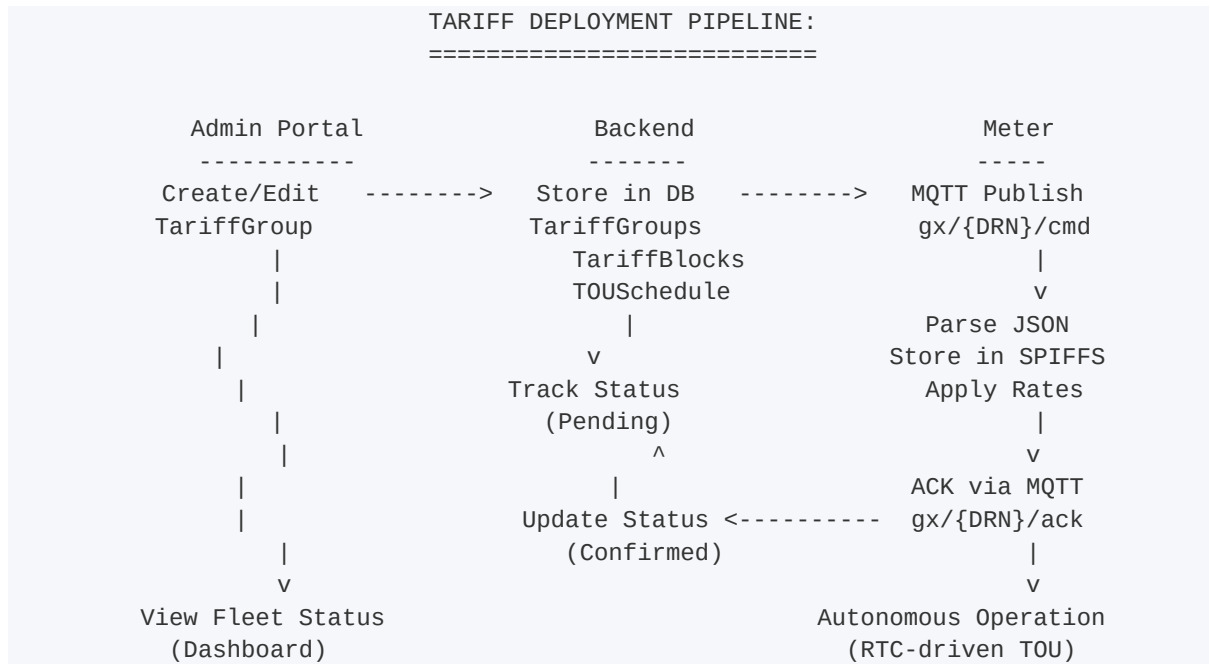
9.1 High-Level Architecture



9.2 Billing Data Flow



9.3 Tariff Synchronisation Flow



9.4 External Integration API

The GRIDx platform provides a RESTful API gateway for third-party integration:

Feature	Endpoint	Description
Token Vending	POST /external/vend	Generate token with full breakdown
Balance Inquiry	GET /external/balance/{meter}	Check arrears and account status
Transaction Status	GET /external/status/{ref}	Query transaction by reference
Webhook Events	POST {webhook_url}	Real-time event notifications

Partners are managed with API keys, rate limiting (60/min default), sandbox/production modes, and granular permissions (vend, balance, status).

10. Recommendations for Improvement

Based on the analysis of the current GRIDx tariff and billing architecture, the following strategic improvements are recommended to enhance system capabilities and position the platform for utility-grade deployment.

10.1 Dynamic Tariff Pricing

Implement real-time dynamic pricing that adjusts tariff rates based on live grid conditions:

- Grid frequency response: Automatically increase rates when grid frequency drops (high demand), decrease when frequency rises
- Renewable integration: Lower rates during periods of solar/wind surplus to encourage consumption
- Demand-response events: Push temporary rate changes to incentivise load reduction during emergencies
- Market-linked pricing: Connect to spot market prices for large commercial/industrial customers

10.2 AI-Powered Load Optimisation

Leverage machine learning to optimise demand-side management:

- Consumption prediction: Train models on historical meter data to forecast demand 24-48 hours ahead
- Optimal scheduling: Use ML to determine best times for geyser/HVAC load shifting based on tariff and solar forecast
- Anomaly detection: Automatically identify unusual consumption patterns indicating theft or meter faults
- Revenue forecasting: Predict weekly/monthly revenue for utility financial planning

10.3 Solar Forecasting Integration

Integrate weather and irradiance forecasting for net metering:

- Short-term solar production forecasts to optimise export/import decisions
- Battery storage dispatch optimisation (for future battery-enabled meters)
- Virtual power plant aggregation of distributed solar assets

10.4 Revenue Protection & Fraud Detection

Enhance the existing tamper detection system with:

- Statistical analysis of consumption patterns across neighbouring meters
- Balance-of-energy calculations at transformer level to detect unmetered losses
- Machine learning classifiers trained on confirmed tamper cases
- Automated investigation workflows with evidence collection

10.5 Smart Prepaid/Postpaid Hybrid Model

Introduce a hybrid billing model that combines the best of both worlds:

- Customers can top up prepaid credit while maintaining a postpaid account
- Automatic fallback to postpaid when prepaid credit is exhausted (no disconnection)
- Credit limits managed based on customer payment history and risk score
- Seamless transition without meter reconfiguration or relay toggling

10.6 Utility-Grade Reporting

Expand reporting capabilities to meet utility regulatory requirements:

- NERSA/ECB regulatory compliance reports (annual, quarterly)
- Loss calculations: Technical losses (transformer level) vs. commercial losses (theft/metering errors)
- Revenue assurance dashboards with KPI tracking
- Customer segmentation analytics for tariff optimisation
- Export-ready reports in PDF, Excel, and CSV formats

11. Conclusion

The GRIDx Smart Metering Platform represents a comprehensive, production-ready tariff management and billing solution designed specifically for the Namibian and SADC electricity market. With support for 27 ECB-compliant tariff categories, dual billing modes, real-time MQTT-based meter communication, and full IEC 62055-41 STS compliance, the platform is uniquely positioned to serve utilities, municipalities, and energy distributors at scale.

The system's architecture ensures operational resilience through autonomous meter-side tariff enforcement, offline capability, and real-time synchronisation. The billing engine's atomic transaction processing and complete audit trail provide the financial transparency required by regulators and utility operators.

With the recommended improvements in dynamic pricing, AI-powered optimisation, and enhanced fraud detection, GRIDx can evolve into a next-generation smart grid management platform that sets the standard for African electricity infrastructure modernisation.

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